TASK SPECIFICATION: P2-04 PHASE P2 — IDENTITY, SECURITY & CAMPUS CORE

User Profile & Vehicle Registration Management API

Git Branch: `feature/P2-04-profile-vehicles-api` · Target Pull Request: feature/P2-04-profile-vehicles-api

Core Objective & Technical Scope:
Self-service API enabling authenticated students and faculty to retrieve and update profiles and register, update, and manage campus vehicles with IDOR protection.

Priority Level	Medium	Complexity Rating	`M`
Execution Status	In Progress ⏳	Target PR / Commit	`feature/P2-04-profile-vehicles-api`
Upstream Dependencies	P2-03		
Architecture Blockers	None (Ready for execution)		

IMPLEMENTATION ROADMAP & ENGINEERING STEPS

1. Implement ProfileService and VehicleService application logic.
2. Create MeController and VehicleController with DTO request/response contracts.
3. Enforce security ownership checks ensuring users can access only their own vehicles.
4. Author comprehensive unit and MockMvc integration tests.

TARGET FILE INVENTORY & EXPECTED ARTIFACTS

- `identity/api/MeController.java`
- `identity/application/*`
- `DTOs`
- `tests`

ACCEPTANCE CRITERIA & QUALITY CHECKLIST

- Users can view and edit only their own profile and vehicles (zero IDOR vulnerability).
- Validation errors emit standard RFC 7807 problem details.

VERIFICATION & TEST COVERAGE STRATEGY

- ProfileServiceTest
- VehicleControllerTest
- IdorSecurityTest